

Article

Polarization Analysis of Vertically Etched Lithium Niobate-on-Insulator (LNOI) Devices

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Abstract: LNOI devices have emerged as prominent contributors to photonic integrated circuits (PICs), benefiting from their outstanding performance in electro-optics, acousto-optics, nonlinear optics, etc. Due to the physical properties and current etching technologies of LiNbO₃, slanted sidewalls are typically formed in LNOI waveguides, causing polarization-related mode hybridization and crosstalk. Despite the low losses achieved with fabrication advancements in LNOI, such mode hybridization and crosstalk still significantly limit the device performance by introducing polarization-related losses. In this paper, we propose a vertically etched LNOI construction. By improving the geometrical symmetry in the waveguides, vertical sidewalls could adequately mitigate mode hybridization in common waveguide cross sections. Taking tapers and bends as representatives of PIC components, we then conducted theoretical modeling and simulations, which showed that vertical etching effectively exempts devices from polarization-related mode crosstalk. This not only improves the polarization purity and input mode transmittance but also enables lower polarization-related losses within more compact structures. As a demonstration of fabrication feasibility, we innovatively proposed a two-step fabrication technique, and successfully fabricated waveguides with vertical sidewalls. Such vertical etching technology facilitates the development of next-generation high-speed modulators, nonlinear optical devices, and other advanced photonic devices with lower losses and a smaller footprint, driving further innovations in both academic research and industrial applications.

Keywords: lithium niobate-on-insulator; vertical etching; mode hybridization; polarization-related mode crosstalk



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1. Introduction

The LNOI platform has emerged as a highly promising candidate for integrated photonics [1,2] due to its favorable material properties, including a high refractive index contrast, wide transparency window, large electro-optic (EO), acousto-optic, and second-order nonlinear coefficients, etc. [3–6]. Recent advancements in fabrication techniques have led to the development of low-loss waveguides and high-Q resonators on the LNOI platform [2,4,7]. For instance, ultra-low loss waveguides with propagation losses as low as 2.7 dB/m and microring resonators with intrinsic quality factors up to 10⁷ have been achieved [8]. Platform advantages along with fabrication progress facilitate the creation of highly efficient and compact photonic devices for a wide range of advanced applications, ranging from high-speed modulators [9–12] to nonlinear optics [13–15], which are comparable to other platforms like Si and GeSn [16,17]. The EO interaction strength is significantly

enhanced in the LNOI platform due to its low losses, close electrode spacing, strong field confinement and high-Q resonators. These attributes enable a range of applications including EO modulation, EO frequency combs, and frequency shifting in the GHz range [6]. One critical aspect of LNOI waveguides is that the nature of their guided optical modes are inherently hybrid, i.e., quasi-transverse electric (qTE) and quasi-transverse magnetic (qTM) modes. For example, a qTE mode exhibits a dominant electric field component along with a non-dominant magnetic field component in one of the transverse directions [18]. Generally, asymmetry in geometry and refractive indices enhances modal hybridness [19]. Severe mode hybridization disrupts the inter-mode orthogonality in the waveguide, leading to overlaps among eigenmodes. Such overlaps cause unwanted coupling between modes of orthogonal polarizations, thereby introducing polarization-related mode crosstalk in devices [20]. Sections of the device with high modal hybridness act as “bridges” for inter-mode crosstalk during propagation [21]. Remarkable mode crosstalk has been seen in common PIC components such as bends [22,23], tapers [19,24–27], and multiplexers [28,29]. Such crosstalk appears as an undesirable polarization-related loss, complicating alternative approaches to maintain a stable polarization state. Significant efforts have been devoted to suppressing inter-mode crosstalk and, hence, preserving the polarization state, including advanced fabrication methods [30] and introducing additional mode converters [31], various types of bends (polynomial, Bézier and Euler curves, and their combinations with arcs) [32–37], and abrupt junctions [38].

Among all sources of structural asymmetry, slanted sidewalls present a significant challenge in exacerbating mode hybridization and crosstalk. The material properties of LiNbO₃, such as its high resistance to etching and the tendency to produce excessive etching byproducts, lead to the unavoidable formation of sidewall slopes [39]. LNOI waveguides fabricated using dry etching techniques typically exhibit sidewall angles ranging from 60° to 75° [40], whereas wet etching techniques result in even smaller angles and are further hampered by orientation-selective anisotropic etching rates [41]. Mode hybridness, increased by structural asymmetry from slanted sidewalls, leads to greater overlap between qTE and qTM modes, which exacerbates mode crosstalk and results in polarization-related losses. These losses impact overall device performance, affecting crucial applications such as high-speed modulation and nonlinear optics.

In this paper, we propose vertical etching structures on LNOI platforms to mitigate the mode hybridization and crosstalk caused by slanted sidewalls. Theoretical analyses show that vertical sidewalls significantly reduce mode hybridization in waveguides when two modes have similar group velocities. Advancing further, the rectangular LiNbO₃ cross section surrounded by SiO₂ cladding effectively decouples qTE and qTM modes. We simulated two commonly used structures in LNOI devices—tapers and bends—to examine the inhibition of polarization-related mode crosstalk in vertically etched waveguides compared to traditional slanted sidewalls. The results imply that vertical etching could effectively enhance polarization purity and mode fidelity in the output ports, which produces a great reduction in polarization-related losses. With vertical sidewalls, moreover, bends can be designed to be more compact to downsize photonic devices. As a demonstration of the feasibility of our theoretical approach, we implemented a practical vertical etching technique. Alternating steps of conventional vertical plasma incidence and inclined plasma incidence were implemented, generating a rectangular 300 nm deep rib waveguide on 600 nm thick LiNbO₃. This fabrication capability may improve geometrical and index symmetry in LNOI, depicting a promising future for highly compact LNOI platforms. The potential for reduced losses and increased device compactness will drive further innovations in both academic research and industrial applications.

2. Simulation

Mode hybridization is intensified when two orthogonally polarized modes that are close in their effective indices couple to each other [19]. Any factors that enhance the non-

dominant field component will physically exacerbate mode hybridization. In high-contrast integrated photonic platforms, intuitively, these factors include the following:

- Equal height and width of the waveguide;
- Geometrical asymmetry in the horizontal or vertical direction;
- Index asymmetry in the vertical direction;
- Asymmetry in the propagation direction (e.g., bends, tapers, etc.).

Common LNOI waveguide configurations avoid near-square cross sections, and SiO₂ cladding can be deposited to achieve index symmetry. However, the issue of geometrical asymmetry introduced by slanted sidewalls during etching remains unavoidable. Strong mode hybridization excited by slanted sidewalls further increases the overlap between qTE and qTM modes. Mode crosstalk will be accumulated when highly hybrid modes propagate through diversified nanophotonic devices containing bends, tapers, and couplers. In this section, we select tapers and bends to analyze the impact of etching angles on mode hybridization and crosstalk, which are inherently present in these structures.

2.1. Mode Hybridization

In order to understand and quantify the impact of etching angles on mode hybridization, it is crucial to simulate and analyze the influence of various geometric and material parameters. Specifically, factors such as the sidewall angle, waveguide top width (TW), and etching depth (ED, or equivalently, slab thickness) play significant roles in determining mode hybridness. A typical cross section of LNOI waveguides is illustrated in Figure 1a. To isolate the impact of slanted sidewall angles on mode hybridization, the LNOI waveguide was embedded in a SiO₂ buried oxide (BOX) layer and top cladding, thereby eliminating index asymmetry. The film thickness was fixed at 600 nm, and the base angle θ was set to 70° and 90° for comparative purposes. Figure 1b depicts the special case where the ED is equal to the thickness (fully etched-through) and $\theta = 90^\circ$ (vertical etching), which maximizes the cross-section geometrical and index symmetry.

To quantify mode hybridization, we need to define the TE polarization fraction γ_{TE} as [19,26,42]

$$\gamma_{TE} = \frac{\iint |E_x|^2 dx dy}{\iint (|E_x|^2 + |E_y|^2) dx dy} \quad (1)$$

where the coordinate system is defined in Figure 1a. Similarly, the TM polarization fraction γ_{TM} is defined as

$$\gamma_{TM} = \frac{\iint |E_y|^2 dx dy}{\iint (|E_x|^2 + |E_y|^2) dx dy} \quad (2)$$

We used the finite element method to obtain the eigenmode fields of the waveguide structures [43]. In the simulation, the TW and ED were varied across common values in devices, creating a detailed map of mode hybridization, as displayed in Figure 1c–h. The ED was scanned from 300 nm to 600 nm and the TW was scanned from 1.2 μm to 4.1 μm . Nonlinear phase matching efficiency is optimized when utilizing the d_{33} coefficient of LiNbO₃ with mode overlap maximized. This construction corresponds to the TE₀₀ mode in *x*-cut or the TM₀₀ mode in *z*-cut LNOI [44]. Two typical wavelengths: 780 nm and 1550 nm, were selected for simulation because they are widely used in nonlinear frequency conversion on LNOI platforms. Figure 1c,f are the γ_{TE} maps for the 780 nm TE₀₀ mode in *x*-cut LNOI, highlighting the differences in mode hybridization when $\theta = 70^\circ$ and 90° . The dark strips in Figure 1c indicate that mode hybridization occurs at any ED with slanted sidewalls. By contrast, in Figure 1f, the strip terminates at a certain ED, indicating little mode hybridization in a near symmetric configuration. When the ED is small (e.g., 300–450 nm), mode hybridization is still obvious since the thick slab layer breaks the vertical geometrical symmetry. Similarly, Figure 1d,g compare the γ_{TM} maps for the 780 nm TM₀₀ mode in *z*-cut LNOI when $\theta = 70^\circ$ and 90° , respectively. Severe mode

hybridization can be seen in Figure 1e,h, which display the γ_{TM} maps for the 1550 nm TM_{00} mode in the z-cut LNOI waveguide with $\theta = 70^\circ$ and $\theta = 90^\circ$, respectively.

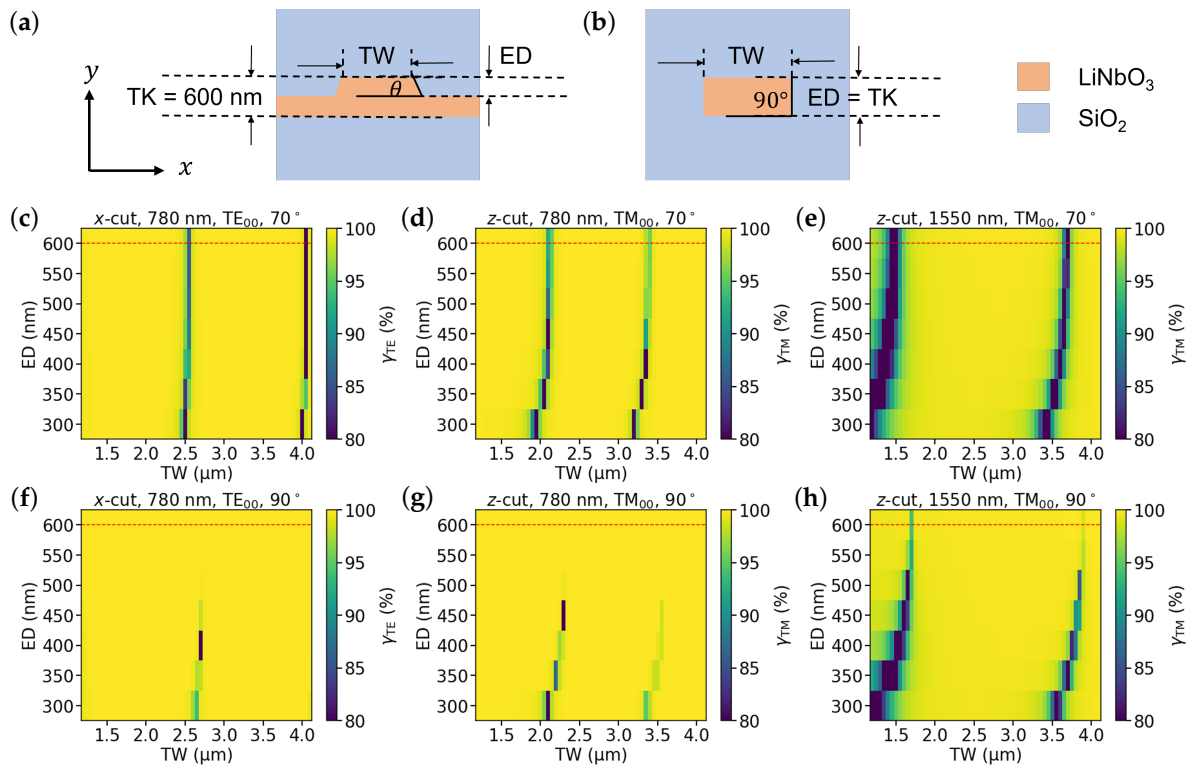


Figure 1. Schematics and simulation results for mode hybridization in rib waveguides with varying etching depths (EDs) and top widths (TWs). (a) The rib waveguide profile in our simulation. Thickness (TK) = 600 nm with a varied ED. The upper strip is an isosceles trapezoid, whose TW and base angle (θ) are defined. (b) When $ED = TK$ and $\theta = 90^\circ$, the rib waveguide degrades into a strip waveguide with a rectangular profile. (c–h) γ_{TE} or γ_{TM} maps when scanning TW and ED. The dark strips indicate severe mode hybridization. (c,f) γ_{TE} maps for the 780 nm TE_{00} mode in the x-cut LNOI waveguide where (c) $\theta = 70^\circ$ and (f) $\theta = 90^\circ$, respectively. (d,g) γ_{TM} maps for the 780 nm TM_{00} mode in the z-cut LNOI waveguide where (d) $\theta = 70^\circ$ and (g) $\theta = 90^\circ$, respectively. (e,h) γ_{TM} maps for the 1550 nm TM_{00} mode in the z-cut LNOI waveguide where (e) $\theta = 70^\circ$ and (h) $\theta = 90^\circ$, respectively.

2.2. Mode Crosstalk in Tapers

Tapers are widely used in nanophotonic waveguides as edge couplers [45–48] and for connecting different units within devices [49–51]. Achieving high-efficiency tapers requires precise modal field matching to ensure compatibility with various waveguide configurations, where a gradient transition of modal field is needed. During the transition, mode crosstalk is physically inevitable. To further investigate the impact of vertical etching on mode crosstalk, we designed a fully etched-through tapered waveguide, as shown in Figure 2. This configuration effectively isolates the influence of geometrical asymmetry caused by slabs on mode crosstalk [52–55]. The 600 nm thick single-layer taper is linearly tapered from a wide input (TW = 5.0 μm) to a narrow output (TW = 1.3 μm) side, making it suitable for use as a broadband edge coupler with a typical dimension. The coordinate system is consistent with Equations (1) and (2) and does not necessarily align with the crystal orientation. The three sets of configurations used in Figure 1 were simulated in this taper configuration, and the results are shown in Figures 3–5.

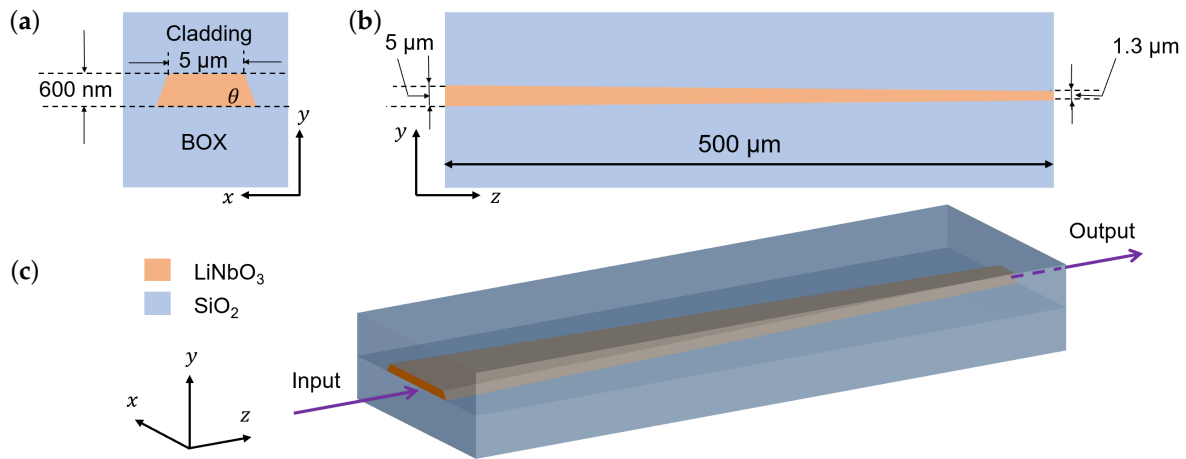


Figure 2. Schematic diagram of the taper in the simulation. The LiNbO₃ waveguide is surrounded by SiO₂ cladding, forming a buried channel waveguide. (a) The side view of the input side shows a 600 nm thick fully etched-through waveguide. TW = 5 μm, and the base angle θ varied in the simulation. (b) The top view shows that the top width is linearly tapered from 5 μm to 1.3 μm in a 500 μm propagating distance. (c) A 3D model of the taper. The beam propagates from the wide side to the narrow side. The coordinate axes indicate the polarization and propagation direction for calculation, which is not consistent with the LiNbO₃ crystal orientation.

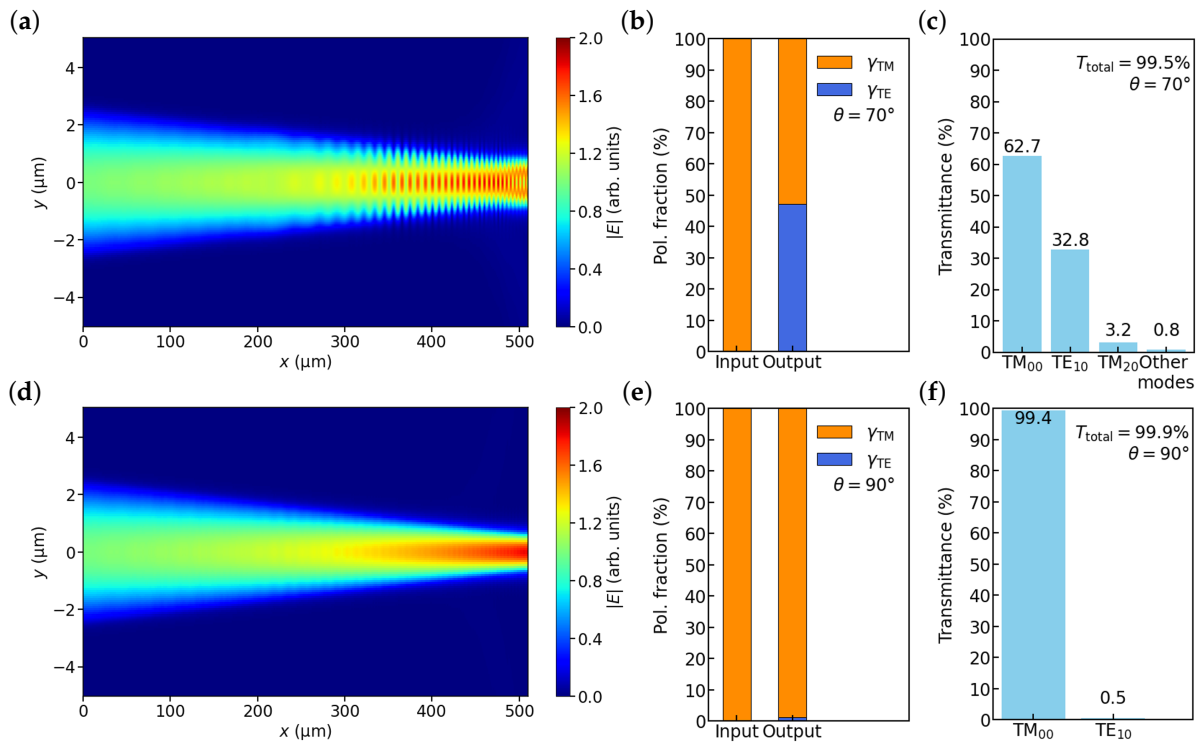


Figure 3. Simulation results of the 1550 nm TM₀₀ mode propagating through the z-cut LNOI tapered waveguide. (a,d) y profile of the input TM₀₀ mode propagating in the taper with base angles (a) 70° and (d) 90°. (b,e) Evolution of γ_{TE} and γ_{TM} from the input TM₀₀ mode to the output field profile where (b) $\theta = 70^\circ$ and (e) $\theta = 90^\circ$. (c,f) Mode components of the output field profile where (c) $\theta = 70^\circ$ and (f) $\theta = 90^\circ$.

Consistent with Figure 1e,h, we found intensified mode crosstalk when the 1550 nm TM₀₀ modes propagate through the z-cut LNOI tapered waveguides with slanted sidewalls. Figure 3 compares such a scenario (Figure 3a–c) to the same mode evolving in a vertically etched taper (Figure 3d–f), with all other taper dimensions being the same. Figure 3a,d show

the y profiles of the field propagating through the taper. We proposed two sets of criteria to characterize the level of crosstalk. Figure 3b,e compare the TM polarization fraction between the input and output fields. A highly hybrid polarization state was observed in the output port. Note that the input TM_{00} modes were intrinsically hybrid modes, although the TM purity was almost 100%. The polarization fidelity was significantly improved during propagation, with γ_{TE} decreasing from 47.3% to 1.2% when the slanted sidewall became vertically etched. However, we were not clear about the mode components that the evolved TE and the left TM fraction belonged to. The second criterion is demonstrated in Figure 3c,f, where the output mode profiles were decomposed into orthogonal components through series expansion. In this way, we could directly obtain the transmittance and conversion of input modes. The transmittance of all modes added up to T_{total} , which indicates the remaining portion of light after the deduction of theoretical insertion/transmission losses. Only 62.7% of the original TM_{00} mode was transmitted, as shown in Figure 3c, compared to 99.4% with vertical sidewalls, as shown in Figure 3f. When $\theta = 70^\circ$, 32.8% of the input TM_{00} mode converted to TE_{10} , 3.2% converted to TM_{20} , and 0.8% converted to higher-order modes. The portion of light that did not maintain its original mode during propagation was considered a polarization-related loss.

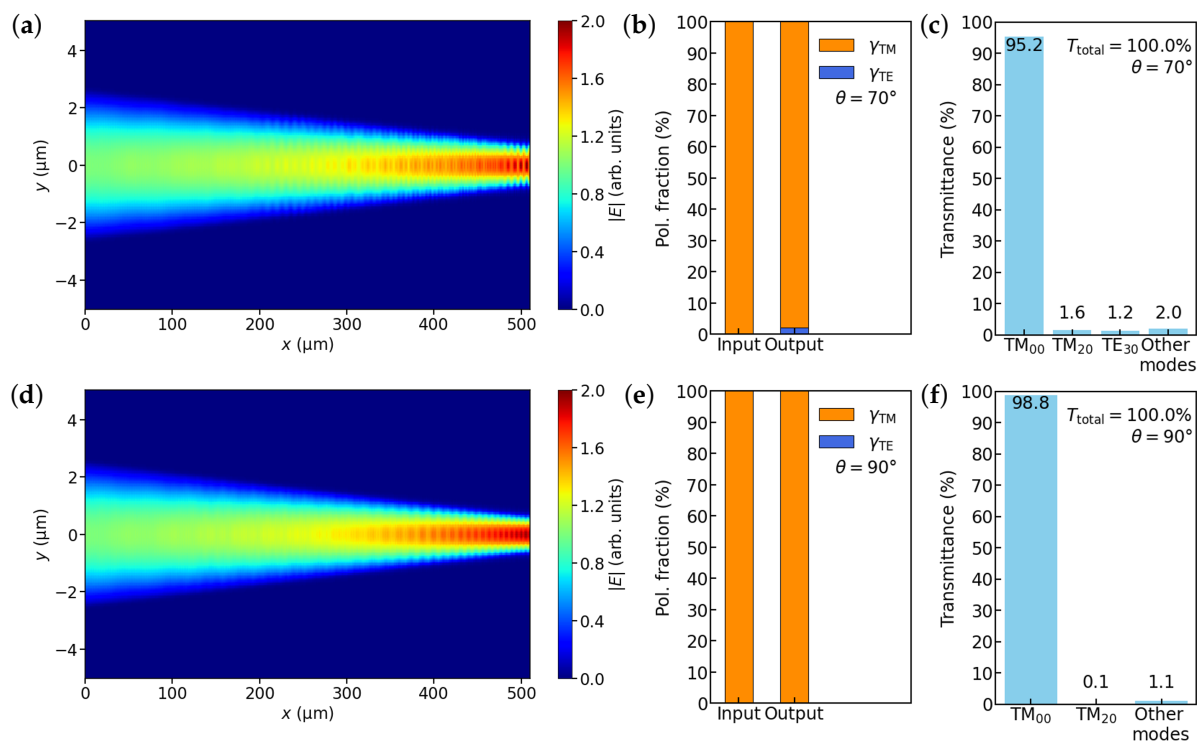


Figure 4. Simulation results of the 780 nm TM_{00} mode propagating through the z-cut LNOI tapered waveguide. (a,d) y profile of the input TM_{00} mode propagating in the taper with base angles (a) 70° and (d) 90° . (b,e) Evolution of γ_{TE} and γ_{TM} from the input TM_{00} mode to the output field profile where (b) $\theta = 70^\circ$ and (e) $\theta = 90^\circ$. (c,f) Mode components of the output field profile where (c) $\theta = 70^\circ$ and (f) $\theta = 90^\circ$.

We then simulated the propagation of 780 nm TM_{00} modes through z-cut LNOI tapered waveguides with both slanted and vertical sidewalls, as shown in Figure 4. Additionally, the propagation of 780 nm TE_{00} modes through x -cut tapered waveguides is illustrated in Figure 5. Similar to Figure 3, the propagation profiles, polarization fraction characterization, and mode transmittance decomposition are illustrated. Only a slight improvement in polarization purity and polarization-related mode fidelity could be achieved in these two examples by vertical etching, because significant accumulation of crosstalk is prevented

when the region of mode hybridization is short and the hybridization is mild, as shown in Figure 1c,d.

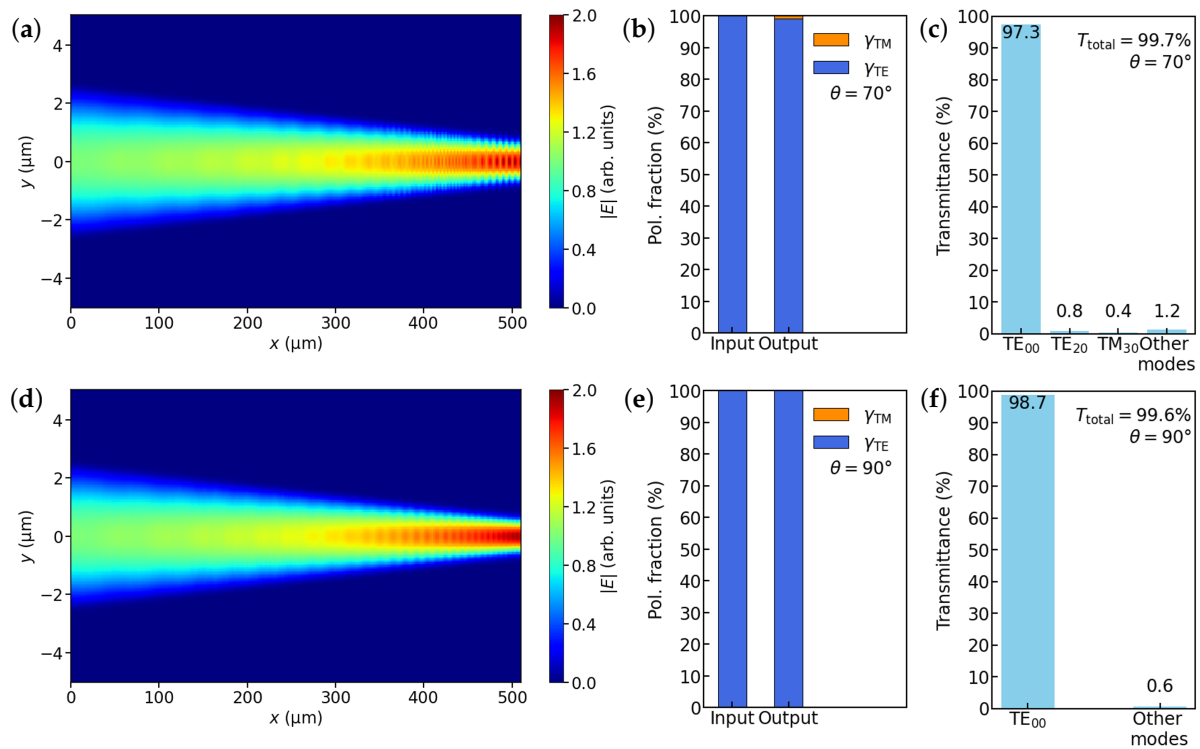


Figure 5. Simulation results of the 780 nm TE_{00} mode propagating through the x-cut LNOI tapered waveguide. (a,d) y profile of the input TE_{00} mode propagating in the taper with base angles (a) 70° and (d) 90° . (b,e) Evolution of γ_{TE} and γ_{TM} from the input TE_{00} mode to the output field profile where (b) $\theta = 70^\circ$ and (e) $\theta = 90^\circ$. (c,f) Mode components of the output field profile where (c) $\theta = 70^\circ$ and (f) $\theta = 90^\circ$.

2.3. Mode Crosstalk in Euler Bends

In addition to tapers, bends are another crucial component in the design of PIC devices. They are commonly used to alter the direction of light propagation, diminish device footprint, and interconnect photonic elements. Besides linear transmission losses, bending losses also include radiative losses arising from a bending mode and polarization-related losses induced by mode crosstalk. To address the issue of bending losses, various bend geometries have been proposed [34], with one of the most popular bends being based on Euler spirals. Euler bends avoid mode coupling to unwanted higher-order modes and reduce radiative losses by linearly increasing the bend curvature along its path length [56]. Nevertheless, even with Euler bends, slanted etching angles still cause crosstalk issues.

Figure 6 delineates a “Euler L-bend” composed of two symmetric 45° Euler spirals [57], which was studied in our simulation. Let α be the turning angle measured from the effective center. When $0 < \alpha \leq \frac{\pi}{4}$, the trajectory equation is written as

$$\begin{cases} x(\alpha) = R_{\text{eff}} \cdot \frac{\int_0^{\sqrt{\alpha}} \cos(t^2) dt}{\int_0^{\sqrt{\frac{\pi}{4}}} [\cos(t^2) + \sin(t^2)] dt} \\ y(\alpha) = R_{\text{eff}} \cdot \frac{\int_0^{\sqrt{\alpha}} \sin(t^2) dt}{\int_0^{\sqrt{\frac{\pi}{4}}} [\cos(t^2) + \sin(t^2)] dt} \end{cases} \quad (3)$$

where R_{eff} is defined as the effective radius of the Euler bend. The minimum radius of curvature turns out at the 45° position. The green dashed line demonstrates a comparison with an arc with the same R_{eff} .

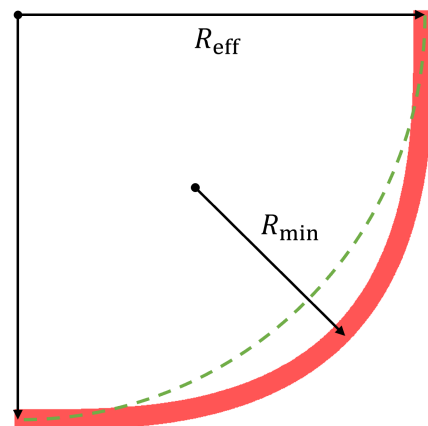


Figure 6. Schematic diagram of a 90° Euler bend. The thick red solid line represents a 45° -to- 45° symmetric Euler bend curve, while the thin green dashed line compares it with an arc of the same effective radius R_{eff} . The minimum radius of curvature, R_{min} , turns out at the 45° position.

Similar to the taper simulations, we attempted to isolate the effects of geometrical and index asymmetry by fully etching thin film LiNbO_3 , cladded with SiO_2 . We studied the polarization fraction evolution and mode transmittance after the beam propagates through the 90° Euler bend in x -cut LNOI, because mode crosstalk becomes almost inevitable in x -cut LNOI where the index ellipsoid inside the transverse cross section varies as the waveguide turns. An example where the 780 nm TE_{00} mode propagates through a Euler bend with $R_{\text{eff}} = 28 \mu\text{m}$ is given in Figure 7, with $\text{TW} = 1.5 \mu\text{m}$. When $\theta = 70^\circ$, there exists a mixture of the 6.4% TM polarization component in the output port, as shown in Figure 7a. Figure 7b then gives the mode spectrum in the output port, indicating significant polarization-related losses as 9.7% of the input TE_{00} mode was converted into higher-order modes. Both polarization fidelity and input mode transmittance were improved with vertical etching, as can be concluded from Figure 7c,d, which implies only a mild deformation of the TE_{00} field profile in the output port. We then modeled the Euler bend structure in Figure 6 with a different R_{eff} . The results of 780 nm and 1550 nm TE_{00} mode propagation are plotted in Figure 8.

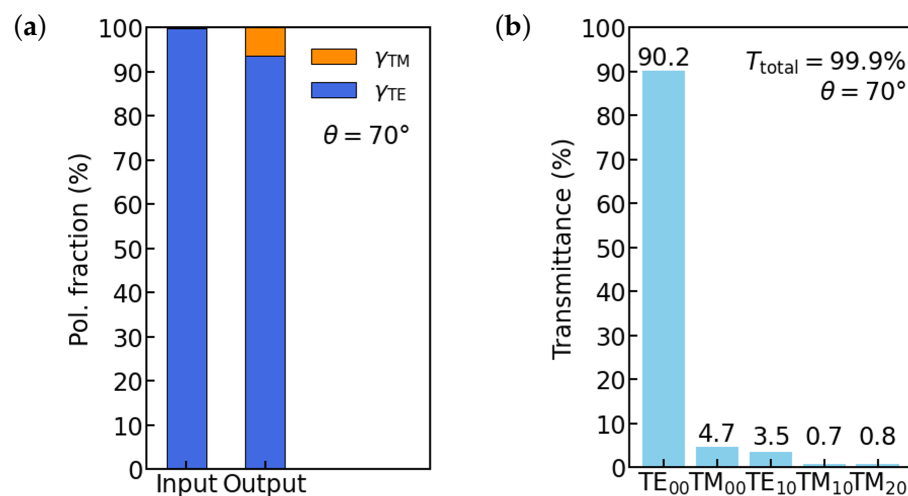


Figure 7. Cont.

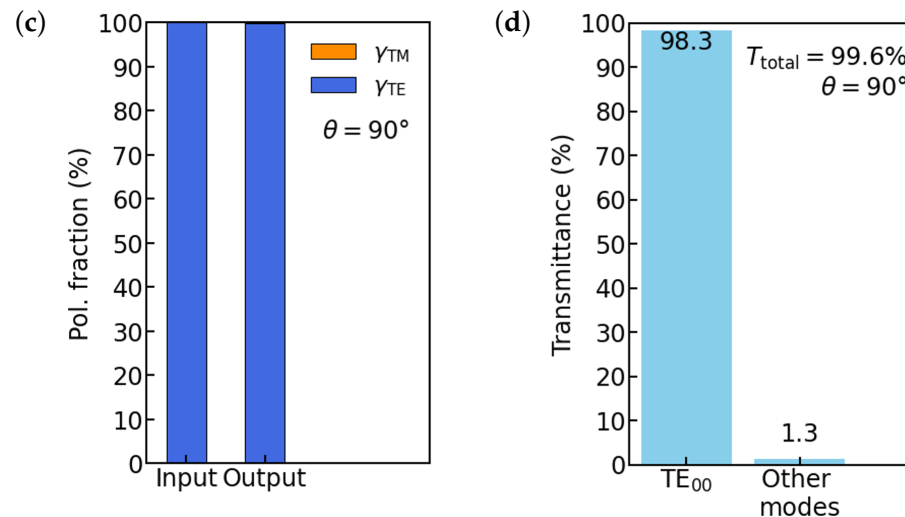


Figure 7. Simulation results of 780 nm TE₀₀ mode propagating through the x-cut LNOI 90° Euler bend composed of two symmetric 45° Euler curves, where $R_{eff} = 28 \mu\text{m}$ and $TW = 1.5 \mu\text{m}$. (a,c) Evolution of γ_{TE} and γ_{TM} from the input TE₀₀ mode to the output field profile where (a) $\theta = 70^\circ$ and (c) $\theta = 90^\circ$. (b,d) Mode components of the output field profile where (b) $\theta = 70^\circ$ and (d) $\theta = 90^\circ$.

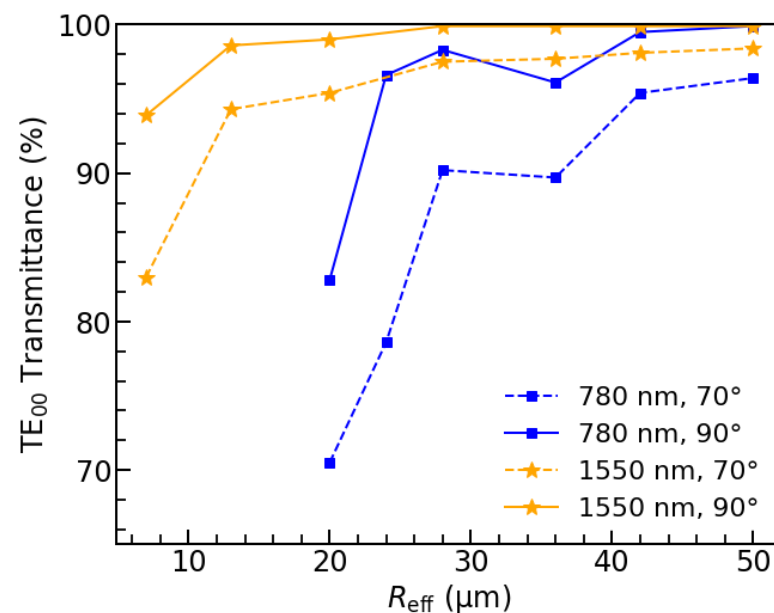


Figure 8. TE₀₀ transmission after propagating through a 90° Euler bend with a TW of $1.5 \mu\text{m}$ versus R_{eff} . Simulated waveguide constructions with base angles 70° (dashed) and 90° (solid) are shown for input TE₀₀ modes with wavelengths of 780 nm (blue) and 1550 nm (orange).

3. Demonstration of Vertical Etching

Besides the simulation, we further examined the feasibility of vertical etching by introducing an innovative two-step dry etching technology. To date, numerous fabrication technique advancements have greatly propelled the development of LNOI photonics [58], including dry etching [59–62], wet etching [63,64], and focused ion beam (FIB) etching techniques [65,66]. Among these techniques, dry etching stands out due to its simplicity and intricacy in fabrication, and compatibility with CMOS processes. The traditional dry etching process involves creating an etching mask on the LNOI surface with high selectivity to LiNbO₃. Subsequently, reactive ion etching, Ar ion milling, or another etching method is employed to etch the LNOI, and finally, the etching mask is removed to produce a rib or ridge waveguide. Despite its advantages of fabricating smooth waveguides

with low propagation losses, sloping sidewalls commonly occur due to the resistance of LiNbO_3 to etching and the excessive generation of byproducts. As etching progresses, these byproducts adhere to the sidewalls, forming a barrier and resulting in a slope.

In our fabrication, the waveguide was patterned using electron beam lithography (EBL) followed by fabrication with an Ar ion milling process [67]. We employed a distinctive method that alternates vertical incidence with inclined incidence in order to achieve vertical sidewalls. By using the inclined plasma incidence, we effectively thinned the sidewall deposits, which improved the verticality of the sidewalls. However, this approach also tended to a notable reduction in etching selectivity. Given the two-step etching process we utilized, a thicker mask was necessary to withstand the additional etching step. We, therefore, meticulously controlled the mask thickness during the experiments, as adequate masking is crucial for maintaining smooth and vertical sidewalls. Throughout the fabrication process, we fine-tuned the etching parameters to achieve optimal results, ultimately producing a waveguide sample with nearly vertical sidewalls. As depicted in Figure 9, we fabricated a waveguide with a 600 nm film TK and an ED of 300 nm. We performed cross-sectional SEM characterization, which revealed the actual sidewall angles of 88.8° and 89.4° , confirming the feasibility of nearly vertical etching. A group of waveguides whose TWs = $1.2 \mu\text{m}$ were fabricated, and the average transmission loss was measured to be 0.69 dB/cm referring to the method in Ref. [8], which is comparable to the conventional waveguide with slanted sidewalls of the same top width. This demonstration, though not fully etched through, underscores the potential of our two-step etching technique for achieving vertical sidewalls.

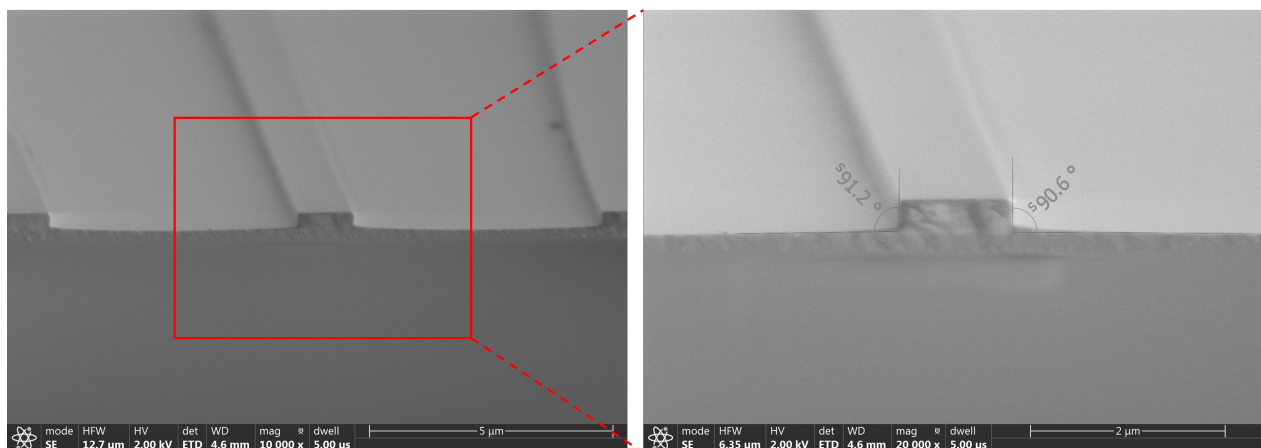


Figure 9. SEM images of the waveguide facet fabricated by the two-step vertical etching. The (left) figure shows that a 300 nm etching depth was achieved on the x-cut 600 nm thick LiNbO_3 to form a rib waveguide. In the zoomed-in image (right), the sidewall angles are nearly vertical.

4. Discussion

Unlike traditional approaches that attempt to avoid crosstalk by selecting specific waveguide configurations, we emphasize the importance of vertical etching to effectively mitigate mode hybridization and crosstalk. Throughout our simulation, the 70° angle represented the standard base angle achieved with advanced dry etching techniques, whereas the 90° angle denoted vertical etching. We first quantified the impact of etching angles on mode hybridization in Section 2.1. When the TW reached a critical value at a fixed ED, two orthogonally polarized modes intersected at their effective refractive indices (n_{eff}) [26]. Hybridization was exacerbated when a large field was tightly confined in a relatively small waveguide cross section even in fully symmetric configurations, as shown in Figure 1e,h. According to Ref. [18], mode hybridization persists when two modes share the same n_{eff} even in the vertically fully etched-through cases. The reason for the absence of observable mode hybridization in Figure 1f,g is that the TW scanning step missed the precise points where the sharp peaks of mode hybridization were theoretically located.

Those “exact structures” are almost non-existent in practice because it is highly improbable for fabricated waveguides to precisely match these configurations. Moreover, fabrication inaccuracies and manufacturing tolerances lead to random deviations from the theoretical design, further reducing the likelihood of encountering these exact structures. These simulations provide insights into how vertical sidewalls in common configurations can mitigate polarization-related issues, serving as a foundational step before examining mode crosstalk in specific components like tapers and bends.

We took the fully etched-through adiabatic tapers and symmetric Euler bends as examples to discuss the amendment of input polarization purity and mode transmittance with vertical sidewalls. In Section 2.2, we characterized mode crosstalk in a tapered waveguide whose TW passes through the hybridization “strips” in Figure 1c–e (the red dashed lines). In Figures 3–5, two complimentary criteria provide a complete insight into the taper performance in terms of polarization and mode fidelity. We can conclude that mode crosstalk intensified when the taper provided ample space for highly hybridized modes to evolve, as shown in Figure 3a–c. This substantial enhancement demonstrates the critical role of vertical etching in maintaining mode fidelity and polarization purity, particularly for long-wavelength TM modes in *z*-cut LNOI waveguides. Our results indicate that vertical etching significantly improves taper performance by reducing mode crosstalk in *z*-cut LNOI tapered waveguides, particularly for long-wavelength TM modes. These findings highlight the critical role of vertical etching in minimizing polarization-related losses and maintaining mode fidelity.

We then studied the polarization fraction evolution and mode transmittance after the beam propagated through the 90° Euler bend in *x*-cut LNOI in Section 2.3. As the modal field continuously changed during bending, larger bending radii and critical TWs were often required to depress mode crosstalk. The comparison between the 1550 nm and 780 nm lines plotted in Figure 8 tells us that mode crosstalk at shorter wavelengths is more severe under the same R_{eff} . This observation is consistent with the physical essence of mode crosstalk, because bends with the same TW and curvature can accommodate more modes and are, therefore, prone to crosstalk. Subsequently, the reduction in polarization-related losses via vertical etching is more pronounced at short wavelengths. Therefore, we can further expect that high polarization and mode fidelity can be achieved for higher-order mode propagation. When we focused on the same wavelength, slanted sidewalls exhibited inferior performance compared to vertical sidewalls. Vertical etching could offset the adverse effects of reduced R_{eff} on mode crosstalk. For example, the 1550 nm TE₀₀ mode transmittance of $R_{\text{eff}} = 13 \mu\text{m}$ fabricated by vertical etching even transcended that of $R_{\text{eff}} = 50 \mu\text{m}$ when $\theta = 70^\circ$. As a consequence, vertical etching exhibited superior performance over traditional slanted sidewalls in the same bending structure for both short and long wavelength modes, because it maximizes geometrical symmetry, thereby reducing mode hybridization and crosstalk. Practically, vertical etching helps reduce polarization-related losses and minimize device footprint.

To demonstrate the fabrication feasibility, we innovatively proposed a two-step etching technique in Section 3. After alternating steps of conventional etching with the vertical plasma incidence and inclined plasma incidence, we achieved a 300 nm ED with nearly vertical sidewalls on a 600 nm thick thin film (Figure 9). This advancement lays a solid foundation for future innovations in LNOI platforms, promising substantial improvements in device performance.

5. Conclusions

In conclusion, we studied and compared mode hybridization and polarization-related mode crosstalk in various LNOI devices with slanted sidewalls of 70° and vertical sidewalls. The simulation results show that mode hybridization can be adequately mitigated via vertical etching due to the improved geometrical symmetry. Mode crosstalk becomes considerably exacerbated only when the device component offers ample space for highly hybrid modes to overlap and evolve with each other. In a practical sense, polarization-

related losses (conversion to other modes) in tapers and bends can be sufficiently inhibited, and the adverse effects on mode crosstalk from downsizing the bend can be compensated for in both short and long wavelengths. As a demonstration of fabrication feasibility, we innovatively proposed a two-step etching technique, yielding a 300 nm ED with nearly vertical sidewalls on a 600 nm thick thin film. Such vertical etching waveguides pave the way for highly compact and efficient photonic devices. In the future, this innovation holds considerable promise for advancing the capabilities of LNOI platforms. The potential for reduced losses and increased device compactness will undoubtedly drive further innovations, not only in academic research but also in industrial applications, leading to breakthroughs in high-speed modulators, nonlinear optics, and other cutting-edge technologies.

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Abbreviations

The following abbreviations are used in this manuscript:

BOX	Buried oxide
EBL	Electron beam lithography
ED	Etching depth
EO	Electro-optic
LNOI	Lithium niobate-on-insulator
qTE	Quasi-transverse electric
qTM	Quasi-transverse magnetic
SEM	Scanning Electron Microscope
TK	Thickness
TW	Top width

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